RESISTANCE PROFILE, TERBINAFINE RESISTANCE SCREENING AND MALDI-TOF MS IDENTIFICATION OF THE EMERGING PATHOGEN

TRICHOPHYTON INDOTINEAE

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Introduction

- Epidemic of tinea infections in (North) India – mainly *tinea corporis* and *tinea cruris*
- Patients demonstrate highly inflammatory, severe lesions
- High level of resistance to antifungal agent **terbinafine** (TRB)
- Emerging prevalence in neighbouring countries, as well as world-wide

Uhrlaß *et al.*, 2022. *Trichophyton indotineae*-An Emerging Pathogen Causing Recalcitrant Dermatophytoses in India and Worldwide-A Multidimensional Perspective

***Trichophyton mentagrophytes* ITS genotype VIII** was identified as causal agent of these infections

Introduction

T. mentagrophytes ITS genotype VIII or *T. indotineae*?

- Morphologically identical to *T. mentagrophytes*, genetically almost identical to *T. interdigitale*

- But: phylogenetic clustering of the 'Indian' isolates
  ecological, physiological and clinical differences
  high resistance level to TRB

Kano *et al*., 2020:
new species ***T. indotineae***

→ **Correct diagnosis is often difficult**, especially when only using morphology for identification

Introduction

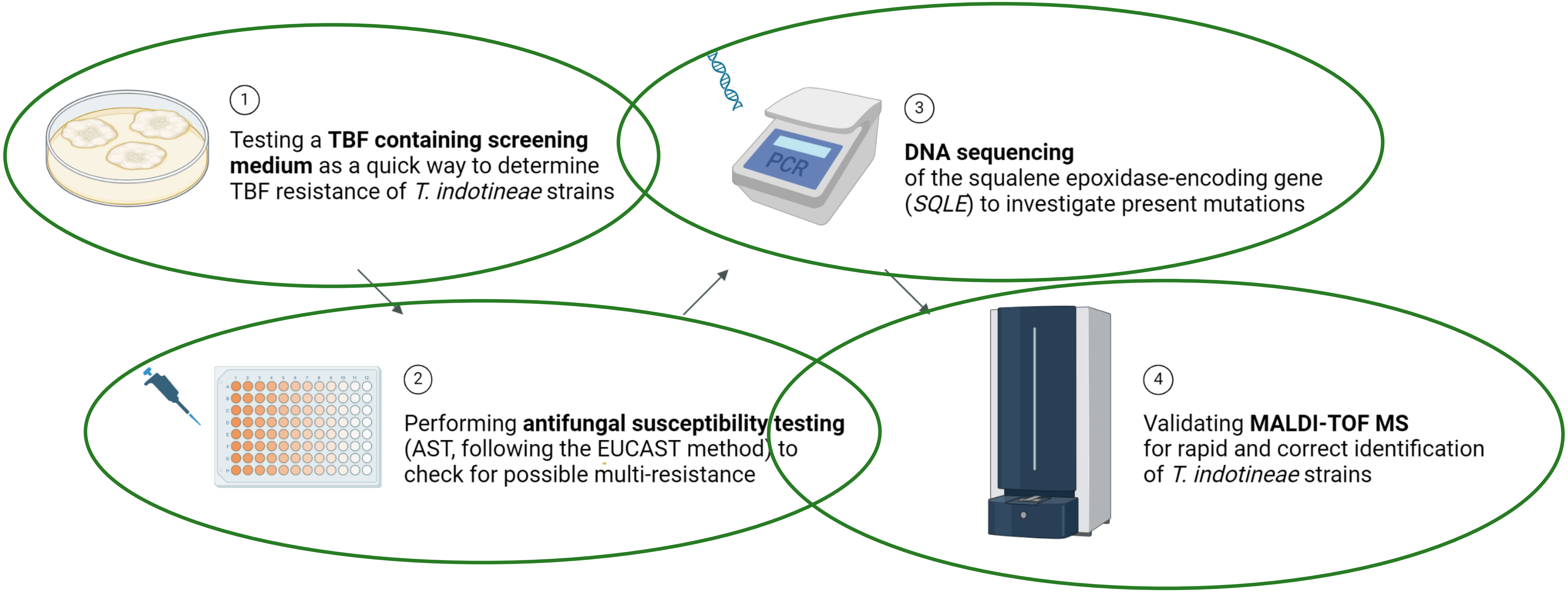
High level of **terbinafine resistance** within *T. indotineae* strains

- Consequence of excessive use of over-the-counter antifungal creams and topical steroids or glucocorticoids?

- Point mutations in the **squalene epoxidase-encoding gene (*SQLE*)**
    - squalene epoxidase (SQLE) is an essential enzyme in the biosynthesis of ergosterol
    - most common amino acid substitutions: **F397L** and **L393F**

- Resistance to other antifungal agents as well?

Research



All strains (n=20) in this study were isolated from the human skin from German patients.
Identity and susceptiblity to TRB was confirmed by Labopart, Leipzig-Mölbis.

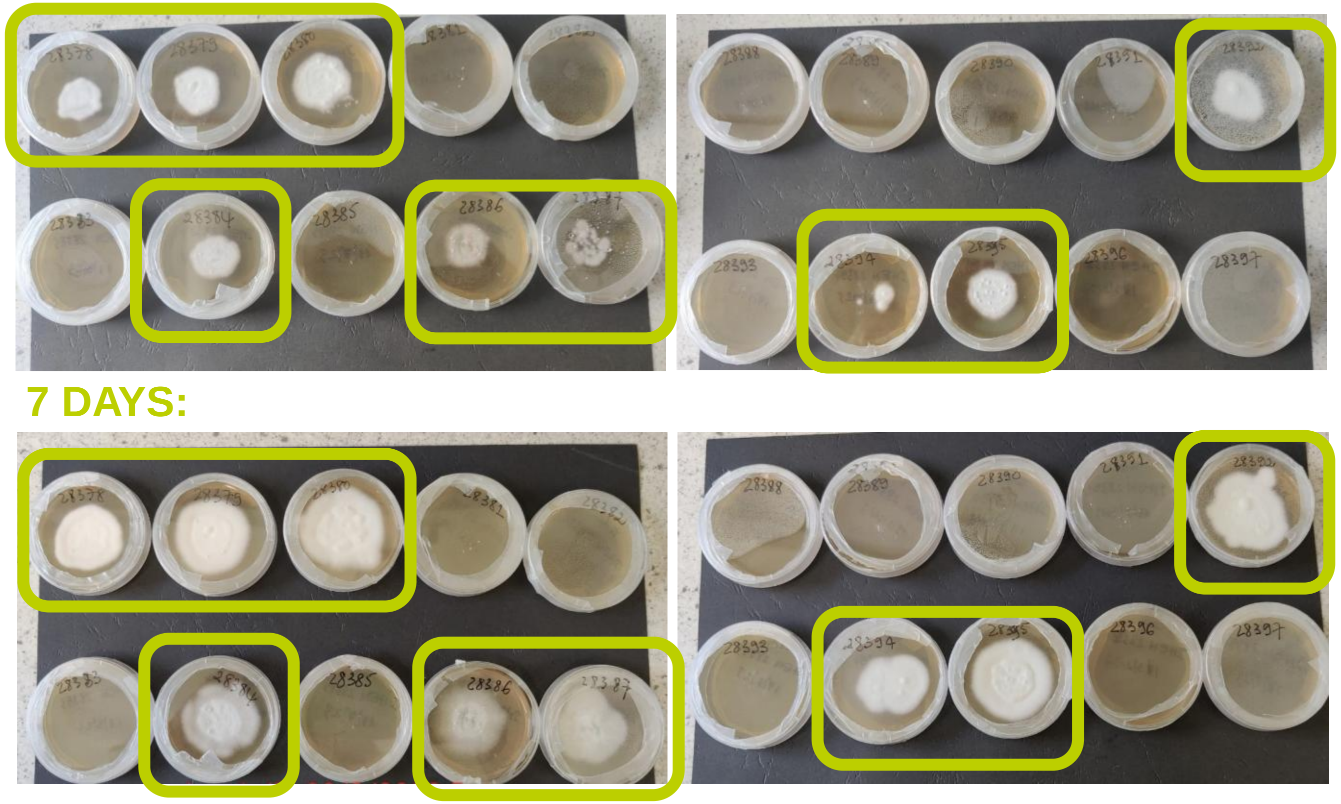
Terbinafine resistance screening

1

Testing a **TBF containing screening medium** as a quick way to determine TBF resistance of *T. indotineae* strains

4 DAYS:

7 DAYS:



- Screening on Sabouraud medium amended with TRB (0.2 µg/mL)
- Incubation of *T. indotineae* strains (n=20) at 25°C
- After **4 days** of incubation: fungal growth of **resistant strains**, while no visible growth for sensitive strains

Figure 1. Growth of the *T. indotineae* isolates (n=20) on a screening medium for terbinafine resistance. The medium consists of Sabouraud dextrose agar, amended with terbinafine.

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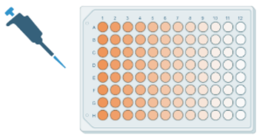
Antifungal susceptibility testing



② Performing **antifungal susceptibility testing** (AST, following the EUCAST method) to check for possible multi-resistance

- EUCAST broth microdilution antifungal susceptibility testing

- Minimal inhibitory concentrations at 100% inhibition of fungal growth (MIC100):

fluconazole	terbinafine
itraconazole	griseofulvin
voriconazole	ciclopirox olamine
ketoconazole	naftifine
	amorolfine

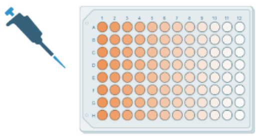
EUCAST EUROPEAN COMMITTEE ON ANTIMICROBIAL SUSCEPTIBILITY TESTING
European Society of Clinical Microbiology and Infectious Diseases

- Tested concentrations ranged from 0.008 – 64 µg/mL

- WT-UL values for itraconazole, voriconazole, terbinafine and amorolfine

  TRB: 0.25 µg/mL, ITC: 1.0 µg/mL, VOR: 2.0 µg/mL and AMO: 1.0 µg/mL

Antifungal susceptibility testing



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Performing **antifungal susceptibility testing** (AST, following the EUCAST method) to check for possible multi-resistance

Table 1. MIC-values (µg/mL) of all T. indotineae strains (n = 20) for tested antifungal agents fluconazole (FLC), itraconazole (ITC), voriconazole (VOR), ketoconazole (KET), terbinafine (TRB), griseofulvin (GRI), ciclopirox olamine (CPX), naftifine (NAF) and amorolfine (AMO).

		MIC (µg/mL)								
***T. indotineae* strains**	**TRB resistance**	**FLC**	**ITC**	**VOR**	**KET**	**TRB**	**GRI**	**CPX**	**NAF**	**AMO**
IHEM 28378	**yes**	16	0.032	0.25	0.25	16	8	1	32	0.25
IHEM 28379	**yes**	32	0.125	0.25	0.25	16	2	1	32	0.125
IHEM 28380	**yes**	64	0.25	0.25	0.5	16	4	1	32	0.125
IHEM 28384	**yes**	16	0,125	0,25	0,25	16	4	1	32	0,125
IHEM 28386	**yes**	32	0,125	0,25	0,25	16	4	0,5	32	0,125
IHEM 28387	**yes**	64	0,25	0,5	1	16	8	0,5	32	0,25
IHEM 28392	**yes**	64	0,25	0,25	0,25	16	4	0,5	32	0,125
IHEM 28394	**yes**	32	0,125	0,25	0,125	16	2	0,5	16	0,25
IHEM 28395	**yes**	32	0.125	0.25	0.25	16	4	0.5	32	0.125
IHEM 28381	**no**	64	0,25	0,25	0,5	0,016	2	1	0,25	0,032
IHEM 28382	**no**	32	0,125	0,125	0,5	0,064	2	1	0,125	0,064
IHEM 28383	**no**	64	0,25	1	1	0,064	4	1	0,25	0,064
IHEM 28385	**no**	32	0,5	0,5	0,5	0,032	4	0,5	0,125	0,064
IHEM 28388	**no**	8	0,125	0,125	0,125	0,064	2	1	0,064	0,064
IHEM 28389	**no**	16	0,125	0,125	0,125	0,125	2	1	0,125	0,125
IHEM 28390	**no**	64	0,25	0,5	0,5	0,125	4	1	0,25	0,125
IHEM 28391	**no**	64	0,25	0,5	0,5	0,032	4	0,5	0,25	0,064
IHEM 28393	**no**	16	0,064	0,125	0,125	0,064	8	0,5	0,5	0,25
IHEM 28396	**no**	32	0.25	0.25	0.5	0.032	2	0.5	0.25	0.25
IHEM 28397	**no**	16	0.064	0.125	0.125	0.032	2	0.5	0.064	0.125

Table 2. Geometric means (GM) of the MIC-values of all *T. indotineae* strains, and separately for the terbinafine (TRB) resistant and susceptible strains. Asterisks (*) symbolize a significant difference between the GM MIC-values of TRB susceptible and resistant strains, after statistical analysis (Mann-Whitney-Wilcoxon test, $p = 0.05$).

	GM MIC-values (µg/mL)		
Antifungal agent	**All (n = 20)**	**TRB susceptible (n = 11)**	**TRB resistant (n = 9)**
Fluconazole	33.13	30.05	37.33
Itraconazole	0.16	0.17	0.15
Voriconazole	0.25	0.25	0.25
Ketoconazole	0.31	0.32	0.29
Terbinafine	0.67	0.05*	> 16.00*
Griseofulvin	3.48	2.92	4.32
Ciclopirox olamine	0.71	0.73	0.68
Naftifine	1.75	0.17*	29.63*
Amorolfine	0.12	0.09*	0.17*

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Antifungal susceptibility testing

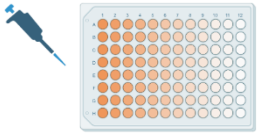

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Table 2. Geometric means (GM) of the MIC-values of all *T. indotineae* strains, and separately for the terbinafine (TRB) resistant and susceptible strains. Asterisks (*) symbolize a significant difference between the GM MIC-values of TRB susceptible and resistant strains, after statistical analysis (Mann-Whitney-Wilcoxon test, p = 0.05).

- Significant increase in geometric mean of the minimal inhibitory concentration (MIC) between TRB susceptible and resistant strains for **naftifine** and **amorolfine**
- Naftifine and terbinafine are both allylamines, while amorolfine is a morpholine antifungal agent
- All strains were considered susceptible for itraconazole, voricazole and amorolfine using WT-UL values by EUCAST
- No significantly elevated MIC values for azoles

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DNA sequencing of *SQLE*

③ **DNA sequencing** of the squalene epoxidase-encoding gene (*SQLE*) to investigate present mutations

known to cause TRB resistance

known to cause azole resistance

		MIC (µg/mL)										
T. indotineae	TRB		ITC	V							*SQLE* missense mutations	amino acid substitution
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	T1189C	F397L
IHEM 28379	**yes**	32	0.125	0.25	0.25	16	2	1	32	0.125	T1189C	F397L
											T1189C	F397L
											T1189C	F397L
											C1191A	F397L
											T1189C	F397L
											C1191A	F397L
											A1179C	L393F
											T1189C & G1342A	F397L & A448T
											G1342A	A448T
											none	none
											G1342A	A448T
											G1342A	A448T
											none	none
											G1342A	A448T
											G1342A	A448T
											G1342A	A448T
											G1342A	A448T
IHEM 28396	**no**	32	0.25	0.25	0.5	0.032	2	0.5	0.25	0.25	G1342A	A448T
IHEM 28397	**no**	16	0.064	0.125	0.125	0.032	2	0.5	0.064	0.125	G1342A & G1330A	A448T

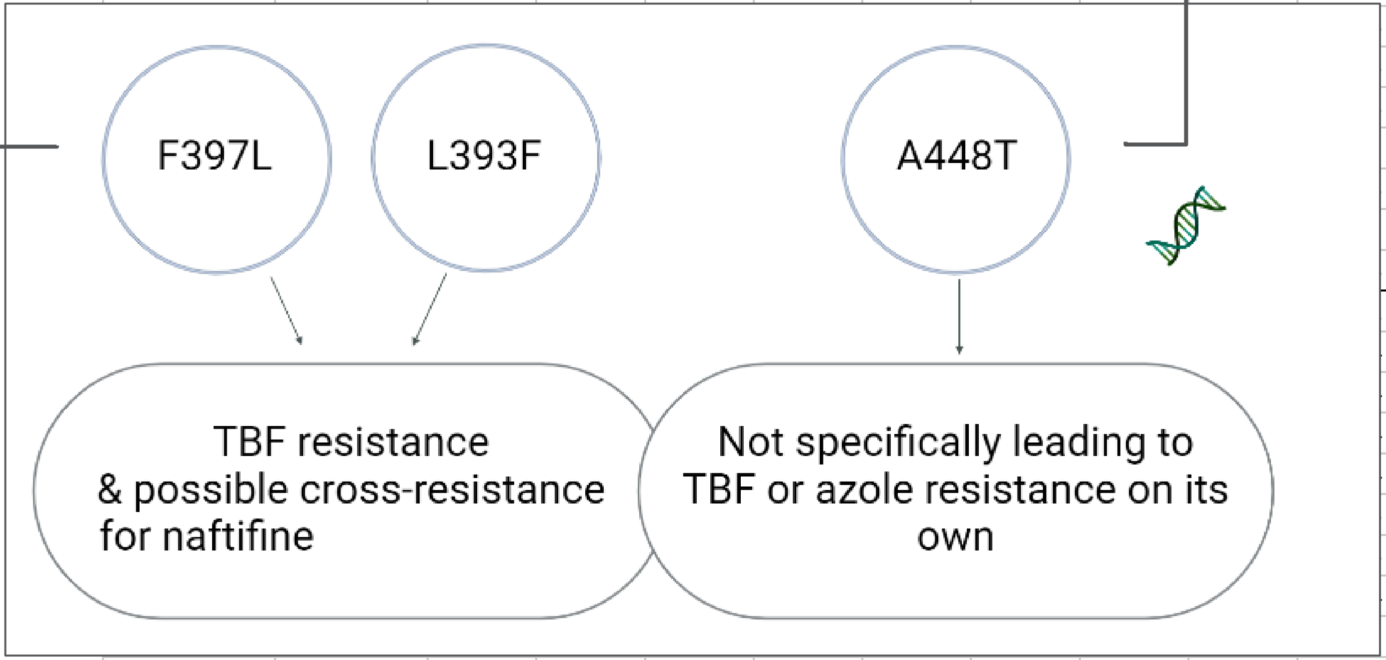
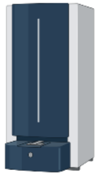


Table 1. MIC-values (µg/mL) of all T. indotineae strains (n = 20) for tested antifungal agents fluconazole (FLC), itraconazole (ITC), voriconazole (VOR), ketoconazole (KET), terbinafine (TRB), griseofulvin (GRI), ciclopirox olamine (CPX), naftifine (NAF) and amorolfine (AMO). MIC-values were obtained by following EUCAST guidelines for the antifungal susceptibility testing of dermatophytes. Missense mutations in *SQLE* and the corresponding amino acid substitutions in the SQLE protein for the tested TRB resistant and susceptible strains. Nucleobases: thymine (T), cytosine (A), adenine (A), guanine (G). Amino acids: phenylalanine (F), leucine (L), alanine (A), threonine (T).

Evaluation of MALDI-TOF MS as an identification tool



4 Validating **MALDI-TOF MS** for rapid and correct identification of *T. indotineae* strains

- Matrix-Assisted Laser Desorption/Ionization Time Of Flight Mass Spectrometry
- Protein extraction based on protocol described by Cassagne *et al.* (2011)
- Spectra were compared to two databases:

1 **in-house database BCCM/IHEM Sciensano**

BCCM BELGIAN CO-ORDINATED COLLECTIONS OF MICRO-ORGANISMS

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2 **MSI V2.0 database**
https://msi.happy-dev.fr/

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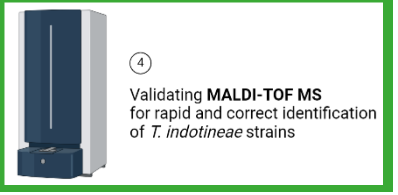
- Can MALDI-TOF MS differentiate TRB resistant and susceptible strains?

  Do susceptible strains have a susceptible reference strain as a best-match and vice versa?

Evaluation of MALDI-TOF MS as an identification tool



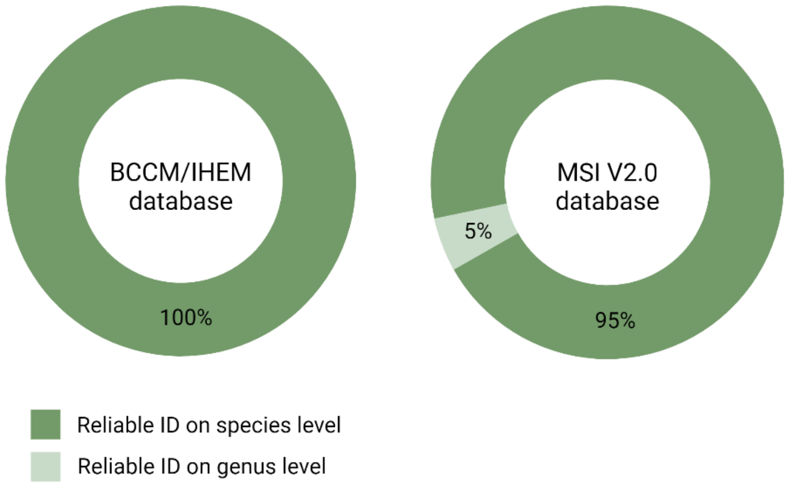
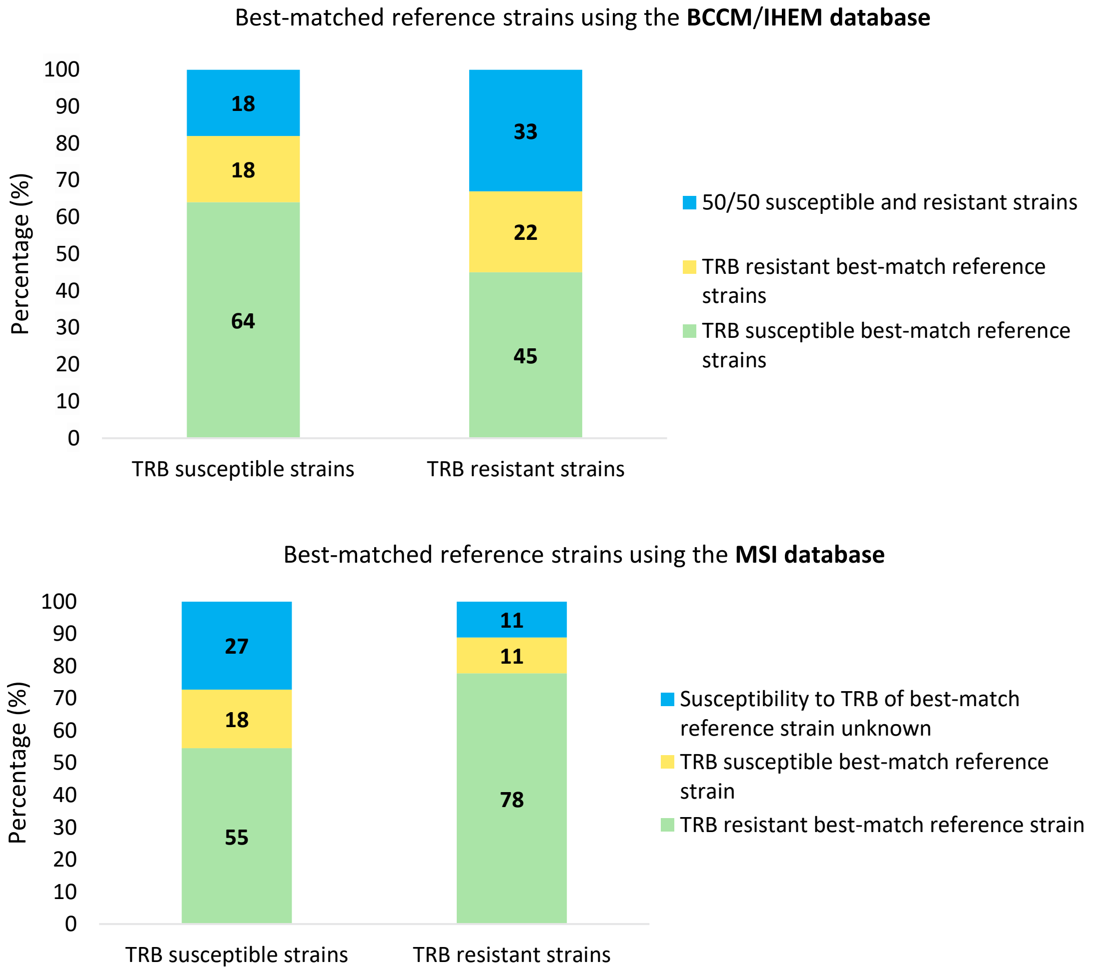


Figure 2. MALDI-TOF MS identification percentages on species and genus level for both databases.



- MALDI-TOF MS can **reliably distinguish** *T. indotineae* strains from closely related species, using both an in-house and public spectral database.

- It could **not differentiate** between TRB susceptible and resistant *T. indotineae* strains.

Figure 3. Column charts of the best-matched reference strains of the *T. indotineae* strains.

Conclusions

- A terbinafine containing Sabouraud medium could screen *T. indotineae* strains for TRB resistance in just four days.
- TRB resistant *T. indotineae* strains showed significantly elevated MIC values for naftifine and amorolfine, raising suspicions of cross-resistance (for naftifine).
- The observed TRB resistance was caused by point mutations in *SQLE*, resulting in amino acid substitutions F397L and L393F, while amino acid substitution A448T does not seem to cause TRB nor azole resistance when no other substitutions are present.
- MALDI-TOF MS is a reliable tool for the identification of *T. indotineae* strains, as it is able to distinguish them from closely related species *T. mentagrophytes* and *T. interdigitale*.
- MALDI-TOF MS could not distinguish TRB resistant *T. indotineae* strains from susceptible ones.

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Dr Renaud Piarroux

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